

MATH 1203 CIS & BIT DISCRETE MATHEMATICS

Lecture 3: COUNTING TECHNIQUES

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Lesson Outcome

By the end of this lesson you should be able to:

- Define the exclusion and inclusion principle as a counting techniques
- Demonstrate understanding and ability to solve problems using exclusion and inclusion principle.
- Describe the Pigeon hole principle
- Apply the Generalized Pigeon hole principle to solve counting problems

INTRODUCTION

- I present to you the third lecture of this course on the first topic, Counting techniques.
- In the previous lecture, you learnt about the Product rule and Sum Rule as a counting technique.
- In this lecture, we will discuss about the Exclusion and Inclusion Principle to solve complex problems.
- We will also discuss about the pigeon hole principle and how we can apply it to solve counting problems.

Exclusion and Inclusion Principle

Definition 1

If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the different ways.

Theorem 2

Suppose A and B are finite sets. Then $A \cup B$ and $A \cap B$ are finite and

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Exclusion and Inclusion Principle CONT'D

- That is , we first find the number of elements in A or B (or both) by first adding $n(A)$ and $n(B)$ (inclusion) .
- And then subtracting $n(A \cap B)$ (exclusion) since its elements were counted twice.

Exclusion and Inclusion Principle CONT'D

Example 3

Suppose a list A contains the 30 students in a Mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists. Find the number of students on list A or B (or both).

Solution

- Since we seek to find $n(A \cup B)$.
- This means we are to combine the two lists and then cross out the 20 names which appear twice.

Exclusion and Inclusion Principle CONT'D

- By Inclusion and Exclusion Principle we have

$$\begin{aligned}n(A \cup B) &= n(A) + n(B) - n(A \cap B). \\&= 30 + 35 - 20 \\&= 45\end{aligned}$$

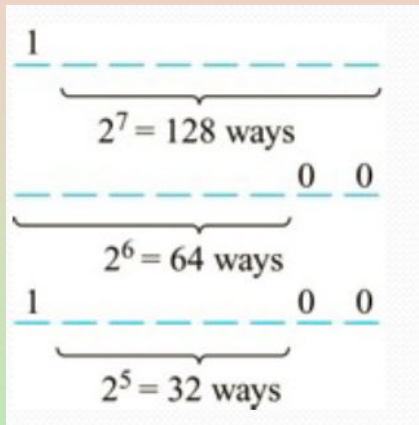
Exclusion and Inclusion Principle CONT'D

Example 4

How many bit strings of length eight either start with a 1 bit or end with bits 00?

Solution

Exclusion and Inclusion Principle CONT'D



- Number of bit strings of length eight that start with 1 bit $= 2^7 = 128$ ways.
- Number of bit strings of length eight that end with bit 00 $= 2^6 = 64$ ways.
- Number of bit strings of length eight that start with 1 bit and end with bits 00 $= 2^5 = 32$ ways.

Exclusion and Inclusion Principle CONT'D

- Hence by exclusion-inclusion principle:

$$\begin{aligned}n(A \cup B) &= n(A) + n(B) - n(A \cap B). \\&= 128 + 64 - 32 \\&= 160 \text{ ways}\end{aligned}$$

Exclusion and Inclusion Principle CONT'D

Theorem 5

For any finite sets A, B, C we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

The Theorem 5 states that we **include** $n(A)$, $n(B)$, $n(C)$, we **exclude** $n(A \cap B)$, $n(A \cap C)$, $n(B \cap C)$ and finally **include** $n(A \cap B \cap C)$.

ACTIVITY

Find the number of Mathematics students at College taking at least one of the Languages, French ,German and Russians, given the following data:

65 study French

45 study German

42 study Russian

20 study French and German

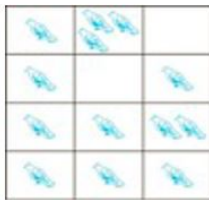
25 study French and Russian

15 study German and Russian

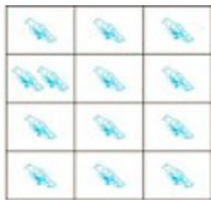
8 study all three languages

Pigeon hole Principle

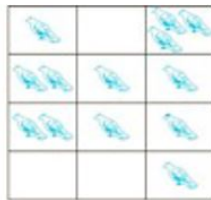
Suppose that a flock of 20 pigeons flies into a set of 19 pigeonholes to roost, then it means that one of the pigeonholes must have more than 1 pigeon. Now let's look on the following diagram which illustrates the pigeon hole rule.



(a)



(b)



(c)

Pigeon hole Principle CONT'D

Theorem 6

The Pigeonhole Principle states that If k is a positive integer and $k + 1$ objects (pigeons) are placed into k boxes (pigeonholes), then at least one box (pigeonhole) contains two or more objects (pigeons).

Example 7

Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Pigeon hole Principle CON'T

Example 8

How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exams is graded on a scale from 0 to 100 points.

Solution

- By counting from 0 to 100
- There are 101 possible scores on the final exams.
- The pigeonhole principle shows that among any 101 students there are at least 2 students with the same score.

Generalised Pigeon hole Principle

Theorem 9

The Generalized Pigeonhole Principle states that If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects

Example 10

Among 100 people , how many were born in the same month?

Solution: Since $N = 100$ and $K = 12$ then the Generalized Pigeonhole Principle tells us that there are at least $\lceil N/k \rceil = \lceil 100/12 \rceil = 9$ who were born in the same month.

Generalised Pigeon hole Principle CON'T

- A common type of Generalized Pigeonhole Principle problems asks for minimum number of objects.
- When we have N objects, the **GPhP** tells us that there must be at least r objects in one of the boxes.
- Therefore the minimum number of objects is the smallest integer $N = K.(r - 1) + 1$ satisfying the inequality $\lceil N/k \rceil \geq r$.

Generalised Pigeon hole Principle CON'T

Example 11

Find the minimum number of students in a class to be sure that at least three of them were born in the same month.

Solution

- Since we have $K = 12$ months and $r = 3$,
- therefore the **GPhP** tells us that the minimum number of students is the smallest integer $N = K(r - 1) + 1$
- Hence among any $12 \cdot (3-1) + 1 = 25$ students, three of them were born in the same month.

Generalised Pigeon hole Principle CON'T

Example 12

Find the minimum number of elements that one needs to take from the set $S = \{1, 2, 3, 4, \dots, 9\}$ to be sure that two of the numbers add up to 10.

Solution

- Here the pigeonholes are the five sets $\{1,9\}$ $\{2,8\}$ $\{3,7\}$ $\{4,6\}$ $\{5\}$.
- Since $K = 5$ and $r = 2$

Generalised Pigeon hole Principle CON'T

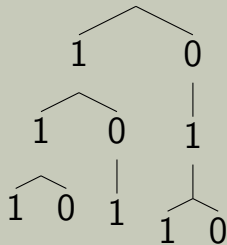
- Then $N = K(r - 1) + 1$ implies $5.(2-1) + 1 = 6$.
- Therefore any choice of six element(pigeons) of S will guarantee that two of the numbers add up to 10.

APPLICATION QUESTIONS

Example

Use a tree diagram to find the number of bit strings of length three with no consecutive 0s.

Solution

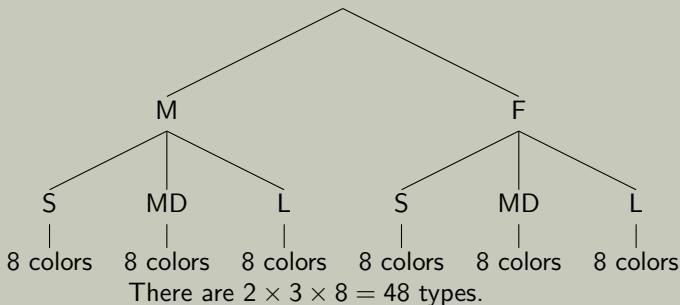


111, 110, 101, 011, 010. So there are 5 bit string of length 3 with no consecutive 0s

Example

A particular brand of shirt comes in 8 colours, has a male version and a female version, and comes in three sizes (small, medium, large) for each sex. How many different types of this shirt are made?

solution



THANK YOU FOR BEING GOOD AUDIENCE!